Two Kinds of Cold Energy: A Capacity-Limited Tethered Component and a Free Core Component Under a Vacuum-Tied MOND Law

Working note

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Abstract

We work with the acceleration law $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ ($\kappa = \frac{1}{2}$, fitted), equivalently $a_0 = \kappa\sqrt{3/8\pi}\, c\, H_\infty \simeq 0.173\, cH_\infty$, where H_∞ is the asymptotic Hubble rate of a cosmological-constant universe. Recent work in the programme wrote the law as the static limit of a single action for a polarization field $\mathbf{P}$ with a saturating vacuum term, $W'(u) = u/(1-u/a_0)$, which needs no new constant and gives $g_N = g^2/(g+a_0)$. That form fits raw SPARC rotation curves at 0.1009/0.0994 dex, and with free-fall basins plus a stiffness rule it passes wide-binary, cluster-satellite and star-node tests.

We report a chain of results that, read from the data first, implies cold energy comes in *two kinds*. (i) The field's bound charge $\nabla\cdot\mathbf{P}$ has zero net mass and no inertia, so it cannot be the collisionless mass that clusters, the Bullet Cluster, Local Group timing and the cosmic web require. (ii) Only a carrier with real rest mass meets the evidence. The viable carrier is a cold collisionless medium whose displacement *is* $\mathbf{P}$ ($\partial_t\mathbf{P} = -\mathbf{j} \Rightarrow \nabla\cdot\mathbf{P} = \rho - \bar\rho$). (iii) That coupling yields a conservative tether whose unique static minimum is the law, with a capacity $\bar\Sigma \le a_0/\pi G$ ($214/258\,\mathrm{M_\odot\,pc^{-2}}$ on our two footings). (iv) In raw SPARC data, galaxies sit below the capacity in the low-acceleration regime (median ≈ 0.3 of it), while the Bullet Cluster's cores hold 3.6–4.8 times it. Cold energy is therefore plausibly *tethered* (law-shaped, capped) in galaxies, with a *free* collisionless component pooling in massive cores. That free component is a candidate for the excess core mass measured in early-type galaxies, groups and 318 eRASS1 clusters. If the medium is a condensate, the data still allow its self-interaction ($\sigma/m \le 0.71\,\mathrm{cm^2\,g^{-1}}$ against a required 0.11–0.82). We list six framework-specific predictions, one decisive measurement, and the open problems, which include the carrier being particle-like and the programme's raw-KiDS shear test still pending.

1 Setting and what is assumed

The acceleration scale is $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ with $\kappa = \frac{1}{2}$ *fitted*. We report two footings, 9.3603×10^{-11} and $1.1312\times10^{-10}\,\mathrm{m\,s^{-2}}$, never pooled. The cold-energy amount $\Omega_c/\Omega_b = 5.364$ is measured, not derived. All numbers come from committed, reproducible lanes of the programme (identifiers CFG n). Several inputs are provisional; they are flagged where used. Verdicts in the programme are judged on raw data, against the framework's own predictions, without ΛCDM-calibrated machinery. Where any such ingredient enters, it is labelled context.

2 The law as one action

The construction (lanes CFG694–696, scored in CFG658) is

$$\mathcal{L} = -\frac{|\nabla\Phi - 4\pi G\,\mathbf{P}|^2}{8\pi G} - \rho_b\Phi - W(\mathbf{P}) - \frac{\gamma}{2}|\nabla\mathbf{P}|^2 + \mathcal{L}_{\rm kin}, \qquad W'(u) = \frac{u}{1-u/a_0}, \tag{1}$$

whose static spherical limit is $g_N = g^2/(g+a_0)$. Each system reads its own free-fall basin, so a uniform external field drops out by the equivalence principle (CFG693). A stiffness rule screens the response below $L_* = 5.75\,\mathrm{pc}$ (CFG691). This rule is not derived from the action.

Results:

- SPARC: 0.1009/0.0994 dex.
- Wide binaries (basins plus stiffness): $\hat\gamma = 1.016/1.020$, -0.8σ against Gaia DR3.
- Cluster satellites: $\chi^2 = 0.7/0.6$ for 5 points.
- A star's own phantom: 0.035 of its mass.

A field's settling motion follows the baryons adiabatically: for realistic assembly times essentially nothing is radiated ($\le 10^{-195}$; CFG659).

3 From data to two kinds of cold energy

(1) The bound charge cannot be the cold mass (CFG722). The field's bound charge has net mass 2.7×10^{-12} of its parts: $+23.9\,M_b$ inside, $-23.9\,M_b$ in an edge shell. It has no inertia of its own and follows the instantaneous field of *all* baryons. In the Bullet Cluster that means the stripped gas, which the merger lanes exclude. The collisionless mass that clusters, the Bullet, Local Group timing and the cosmic web require (CFG624) must therefore be carried by something else.

(2) Only rest mass works (CFG723). We tested the alternatives against data-established properties only:

- the field's energy (it reaches $\sim 10^{-11}$ of the needed density at recombination);
- energy displaced from the vacuum (no inertia; it breaks local energy–momentum balance);
- gravitating-but-inertialess mass (Bullet, $Z = +7.9$);
- coherent field oscillations (Jeans lengths of Mpc).

All fail. A cold collisionless medium works, with $\mathbf{P}$ its displacement. Its mass value may be set by $\rho_{\rm DE}$, but its kind is inertial rest mass. We state plainly that this is particle-like.

(3) A capacity (CFG722). With the medium as the polarizable carrier, the action gives a conservative tether, $C_{\rm sel} = -\nabla_v \cdot (a_T f)$ with $a_T = W'(u)\hat{\mathbf{P}}$. Its unique static minimum is exactly the law (to 3×10^{-16}), and the projected density it can hold is capped:

$$\bar{\Sigma}_{\rm tethered} \leq \frac{a_0}{\pi G} = 214\ (258)\ {\rm M}_\odot\,{\rm pc}^{-2}. \tag{2}$$

This rests on three conditions: the identification itself, a basin-frame anchor, and an export channel the action does not yet contain.

(4) Galaxies below, cluster cores above (CFG724). We used raw SPARC rotmod data for 175 galaxies, with $\Sigma_{\rm dark}(< R) = (V_{\rm obs}^2 - V_{\rm bar}^2)R/(G\pi R^2)$, in the regime $g_{\rm bar} < a_0$. Results (stellar mass-to-light ratios $\Upsilon = 0.5/0.7$):

- 4.6% (canonical) / 3.2% (alt) of points lie above the capacity;
- the median is $63\,{\rm M}_\odot\,{\rm pc}^{-2}$, about 0.3 of the capacity;
- the few $> 3\sigma$ excursions are bulge-dominated spirals, and they vanish at $\Upsilon = 0.8/1.0$.

For the Bullet Cluster we used strong-lensing $M(< 250\,{\rm kpc})$ (provisional) minus the projected baryons. That gives a dark Σ of 1026 and $942\,{\rm M}_\odot\,{\rm pc}^{-2}$ for the two cores, **3.6–4.8 times the capacity**.

Conclusion. If the capacity is real, law-shaped (*tethered*) cold energy can supply at most about a quarter of a massive cluster core's dark mass. The remainder is a *free*, collisionless component that follows the total potential and pools in deep cores. Independent raw-data results show extra core mass exactly there:

- early types: $+0.280 \pm 0.083$ dex at 50–100 kpc, $Z = 3.4$–3.7 (CFG669/658);
- eFEDS groups: more mass beyond the members than either source model (CFG652);
- 318 eRASS1 clusters: $\times 1.8$ inside $\sim$1 Mpc at 7σ, while the outer 1.2–3 Mpc lensing matches the law (CFG687).

The two-population model is being scored against these targets (CFG721). This note does not claim that result.

4 If the medium is a condensate (CFG725)

- **Merger offset.** The offset between galaxies and cold energy scales as $\delta \simeq \eta(\sigma/m)\Sigma_{\rm tgt}V/\omega$, about 41–44 kpc per ${\rm cm}^2\,{\rm g}^{-1}$ at Bullet conditions. The programme's merger estimator gives $\sigma/m \leq 0.71$; reasonable variants give 0.45–10.
- **Window.** A condensate needs 0.15–0.82 (canonical) or 0.11–0.59 (alt). The window is open (partly open on the canonical footing, where Milky-Way-mass halos are marginal).
- **Dwarfs.** Dwarf cores do not test it: they sit below the capacity. One exception, NGC 3738, lies 2–4 times above the capacity and is flagged.

5 Predictions

1. No system's *law-shaped* dark surface density exceeds $a_0/\pi G$; any excess is free cold energy with a non-law profile.
2. Galaxies above the capacity across their whole rotation curve (e.g. NGC 3738) deviate from the radial-acceleration-relation shape in the direction of a free component.
3. The free fraction is $\simeq 0$ in galaxies and $\simeq 0.7$–0.8 in massive cluster cores.
4. **Decisive:** in old, relaxed clusters the free central density is $\rho_{0,\mathrm{free}} \lesssim (1.4\text{–}5.7) \times 10^{-3}\,\mathrm{M_\odot\,pc^{-3}}$ if the medium is a condensate ($\sigma_v = 1000\,\mathrm{km\,s^{-1}}$, $10\,\mathrm{Gyr}$). Measure it with strong lensing plus central-galaxy kinematics, subtracting baryons and the tethered capacity. A denser free core rules out the condensate branch.
5. Merger offsets scale as $\sim$40 kpc per $\mathrm{cm^2\,g^{-1}}$.
6. Direct-detection and laboratory searches are null: the carrier couples only through gravity, and the law is off at Earth's gravity.

Sealed tests elsewhere in the programme:

- a flat $a_0(z)$ for constant ρ_{DE} (ALMA, 2026-11-06 and 11-28);
- Gaia DR4 wide binaries and Milky Way vertical force (2026-12-02), which can falsify the framework but cannot confirm it against ΛCDM.

6 Limitations (plainly)

- The carrier is particle-like. This conflicts with the programme's earlier aim of avoiding a dark-matter particle.
- κ is fitted, Ω_c/Ω_b is measured, and the stiffness length is a rule.
- The capacity rests on the identification, a basin-frame anchor (an unbuilt "switch") and an export channel missing from the action.
- Hot versus condensed baryons is unresolved: the action does not distinguish them, yet restricting the settling source to condensed baryons removed 80% of the programme's raw-KiDS shear tension (CFG596 → CFG636). The full combined shear test is pending.
- Massive early types still fail at $Z \approx 3.7$, and the outer cluster lensing scales with gas mass as $M_{\mathrm{gas}}^{0.90}$ rather than the law's 0.47 (CFG687). Both are open.
- The Bullet lensing masses are provisional, and the two-population model has not yet been scored.
- This is a working note. Nothing here closes the theory.

AI assistance. Anthropic's Claude (analysis, code, text), with contributions from parallel AI research sessions in the same programme. Every number is drawn from committed lanes of the programme's public repository.